disposition after first half-hour

$$\left[360\cdot\frac{2}{3}+360\cdot\frac{1}{2}\;\middle|\;360\cdot\frac{1}{3}+360\cdot\frac{1}{2}\right]\cdot\begin{bmatrix}\frac{2}{3} & \frac{1}{3}\\ \frac{1}{2} & \frac{1}{2}\end{bmatrix}=\text{ transition matrix}$$

$$\left[\left(360\cdot\frac{2}{3}+360\cdot\frac{1}{2}\right)\cdot\frac{2}{3}+\left(360\cdot\frac{1}{3}+360\cdot\frac{1}{2}\right)\cdot\frac{1}{2}\quad\left(360\cdot\frac{2}{3}+360\cdot\frac{1}{2}\right)\cdot\frac{1}{3}+\left(360\cdot\frac{1}{3}+360\cdot\frac{1}{2}\right)\cdot\frac{1}{2}\right]=$$

disposition after second half-hour

$$\left[360\cdot\frac{2}{3}\cdot\frac{2}{3}+360\cdot\frac{1}{2}\cdot\frac{2}{3}+360\cdot\frac{1}{3}\cdot\frac{1}{2}+360\cdot\frac{1}{2}\cdot\frac{1}{2}\;\middle|\;360\cdot\frac{2}{3}\cdot\frac{1}{3}+360\cdot\frac{1}{2}\cdot\frac{1}{3}+360\cdot\frac{1}{3}\cdot\frac{1}{2}+360\cdot\frac{1}{2}\cdot\frac{1}{2}\right]=[430\quad 290]$$